PG No. 25 C_{6v} $6mm$ [hexagonal]

* character table

C_{6v}	$1(1)$	$2_{001}(1)$	$3^+_{001}(2)$	$6^+_{001}(2)$	$m_{100}(3)$	$m_{120}(3)$
A_1	1	1	1	1	1	1
A_2	1	1	1	1	−1	−1
B_1	1	−1	1	−1	1	−1
B_2	1	−1	1	−1	−1	1
E_1	2	−2	−1	1	0	0
E_2	2	2	−1	−1	0	0

* polar ↔ axial conversion

A_1 (A_2) A_2 (A_1) B_1 (B_2) B_2 (B_1) E_1 (E_1) E_2 (E_2)

* symmetric product

	A_1	A_2	B_1	B_2	E_1	E_2
A_1	A_1	A_2	B_1	B_2	E_1	E_2
A_2		A_1	B_2	B_1	E_1	E_2
B_1			A_1	A_2	E_2	E_1
B_2				A_1	E_2	E_1
E_1					$A_1 + E_2$	$B_1 + B_2 + E_1$
E_2						$A_1 + E_2$

* anti-symmetric product

A_1	A_2	B_1	B_2	E_1	E_2
–	–	–	–	A_2	A_2